RAJALAKSHMI ENGINEERING COLLEGE

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Register No.:

Continuous Assessment-I
Computer Science and Engineering
Degree: B. E / B. Tech Semester: III Year: II
Course code & Title: CS23331 & Design and Analysis of Algorithms
Date: 05.09.26 QP CODE: 071103
[Common to: IT, AIML, AI&DS and CSD]

[Regulations: 2023]

Time: 2 hr 15 min Max. Marks: 75

Answer All Questions

Part A: 10 x 2 = 20 Marks

1	Define space complexity. Distinguish between fixed space and variable space requirements.	[CO1]	[K1]
2	Why is the counter method used for determining the time complexity of a non-recursive algorithm?	[CO1]	[K2]
3	Compare Big-O and Big-Θ notations based on the bounds they represent.	[CO1]	[K2]
4	Arrange the following functions in increasing order of growth: log n, n, n log n, n^2, 2^n	[CO1]	[K3]
5	Define a recurrence relation. Why is it used in the analysis of recursive algorithms?	[CO2]	[K1]
6	Identify the applicable case of the Master Theorem for the recurrence $T(n) = 2T(n/2) + n$ and state its asymptotic order.	[CO2]	[K3]
7	Why does Kruskal's algorithm consider edges in non-decreasing order of their weights?	[CO2]	[K2]
8	List the selection criteria used in the Fractional Knapsack Problem and the Activity Selection Problem.	[CO2]	[K1]
9	Recall the three fundamental steps involved in the Divide and Conquer technique.	[CO3]	[K1]
10	Why must the input elements be sorted before applying Binary Search?	[CO3]	[K2]

Part B: 5x11=55 Marks

11 a. Consider the following algorithm: [CO1] [K2]

```
sum ← 0
for i ← 1 to n
    for j ← 1 to i
        sum ← sum + 1
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Identify the basic operation and derive its exact execution count using mathematical summation. Determine the time complexity of the algorithm using asymptotic notation. Illustrate the variation in the number of executions of the inner loop with respect to the outer-loop variable.

[OR]

b. Consider an algorithm whose running time is [CO1] [K2]

$T(n) = 6n^3 + 4n^2 + 8n + 10$

Determine its asymptotic efficiency class. Using the limit method, compare its order of growth with n^2 and n^4, and justify the resulting asymptotic bound. Sketch the relative growth of n^2, n^3, and n^4.

12 a. A travelling salesperson has to visit four cities A, B. C, and D exactly once and return to the starting city. The travel-cost matrix is given below: [CO1] [K3]

	A	B	C	D
A	0	10	15	20
B	10	0	35	25
C	15	35	0	30
D	20	25	30	0

Apply the exhaustive-search technique to generate all feasible tours starting from city A. Calculate the cost of each tour and determine the minimum-cost tour. Also determine the time complexity of the exhaustive-search approach.

[OR]

b. A knapsack has a maximum capacity of 10 kg. Four items have the following weights and profits: [CO1] [K3]

Item	I_1	I_2	I_3	I_4
Weight (kg)	2	3	4	5
Profit	3	4	5	8

Apply exhaustive search to solve the 0/1 Knapsack Problem. Generate all possible subsets, identify the feasible subsets, and determine the subset that produces the maximum profit. State the efficiency of the exhaustive-search approach.

13 a. Consider the recurrence relation [CO2] [K3]

$T(n) = 4T(n/2) + n^2, \quad T(1) = 1$

Solve the recurrence using the Master Theorem and Substitution method. Identify the applicable case of the theorem and determine the asymptotic time complexity.

[OR]

b. Consider the recurrence relation [CO2] [K3]

$T(n) = 2T(n/2) + n, \quad T(1) = 1$

Solve the recurrence using the forward and backward substitution method and establish its asymptotic bound.

14 a. Consider a connected weighted graph with [CO2] [K3]

$V = \{A, B, C, D, E, F\}$

and the following weighted edges:

(A,B,4), (A,F,2), (B,C,6), (B,F,5), (C,D,3), (C,F,1), (D,E,2), (E,F,4)

Apply Kruskal's algorithm to construct the Minimum Spanning Tree. Arrange the edges in non-decreasing order of weight, show the sequence of accepted and rejected edges, and determine the total cost of the Minimum Spanning Tree. Draw the weighted graph and the resulting Minimum Spanning Tree.

[OR]

b. Construct a Huffman code for the following symbols and their frequencies: [CO2] [K3]

Symbol	A	B	C	D	E	F
Frequency	5	9	12	13	16	45

Show each merging step used in constructing the Huffman tree. Assign binary codes to all the symbols and calculate the weighted path length. Draw the complete labelled Huffman tree.

15 a. Apply the Divide and Conquer Min-Max algorithm to the following array: [CO3] [K4]

{35, 12, 78, 4, 56, 91, 23, 67}

Show the recursive decomposition of the array. Determine the minimum and maximum values obtained at each combining stage, identify the final minimum and maximum values, and calculate the total number of comparisons performed. Represent the recursive decomposition using a tree.

[OR]

b. Consider the following array: [CO3] [K4]

{−2, −5, 6, −2, −3, 1, 5, −6}

Apply the Divide and Conquer technique to solve the Maximum Subarray Problem. Determine the maximum subarray by evaluating the maximum left subarray, maximum right subarray, and maximum crossing subarray at the major recursive stages. Identify the resulting maximum subarray and its sum, and determine the time complexity of the algorithm. Represent the recursive decomposition using a tree.